

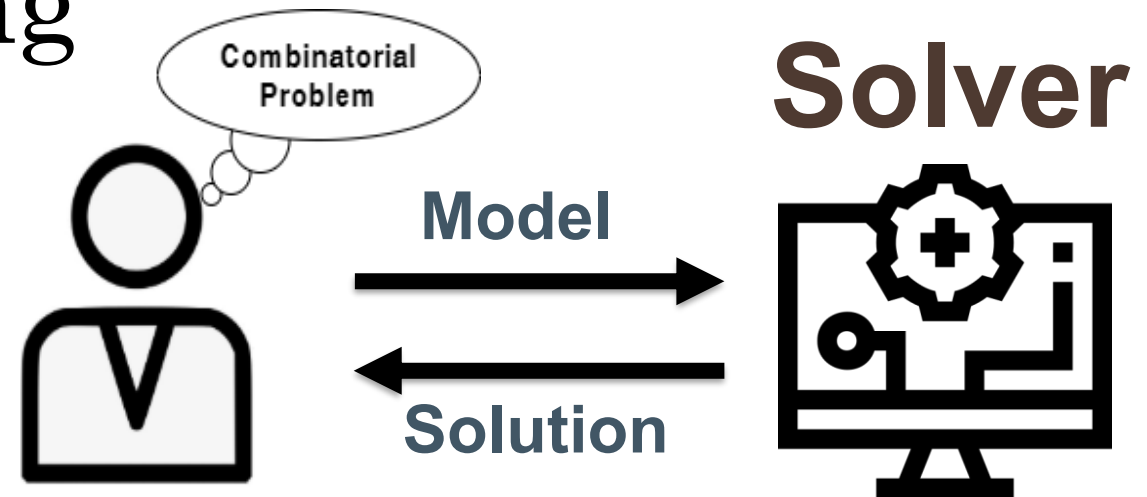
ML-based Generalization Queries for Interactive Constraint Acquisition

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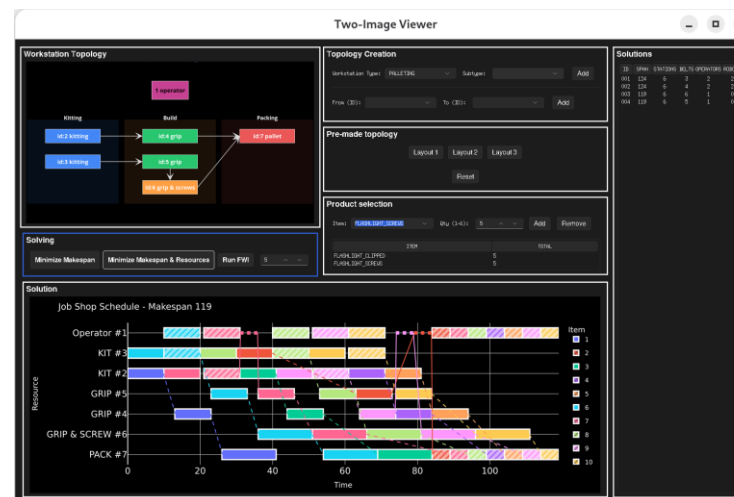
19/07/2026

Constraint Modeling

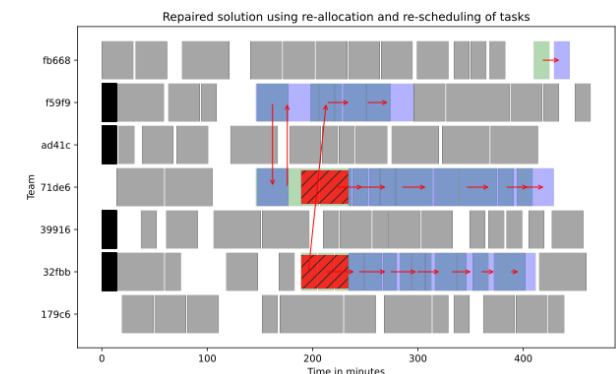
Model + Solve paradigm



- Decision Variables
- Domains
- Constraints
- Objective (optional)



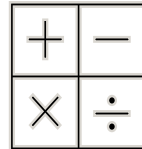
	Week 1							Week 2							Total shifts
	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	
name															
Megan	F	D	D	D	D	F	F	D	D	F	F	D	D	D	9
Katherine	D	D	D	D	D	F	F	D	D	F	F	D	D	F	9
Robert	D	D	D	F	F	D	D	F	F	D	D	F	F	F	8
Jonathan	D	D	F	F	F	D	D	D	D	D	F	F	F	F	7
William	F	D	D	D	D	F	F	D	D	F	D	D	D	D	9
Richard	D	D	D	D	D	F	F	D	D	F	F	F	D	D	9
Kristen	F	F	D	D	D	F	F	D	D	F	F	F	D	D	8
Kevin	D	D	F	F	F	F	F	F	D	D	D	D	D	F	7
Cover D	5/5	7/7	6/6	5/4	5/5	2/5	2/5	6/6	7/7	4/4	2/2	5/5	6/6	4/4	14



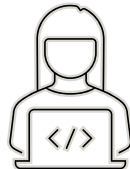
Constraint Solving technologies
are **powerful**, but modeling
requires **expertise** in:



Domain Knowledge



Mathematical Skills



Solver Formalisms

Making Constraint Solving More Accessible

Modelling Tools

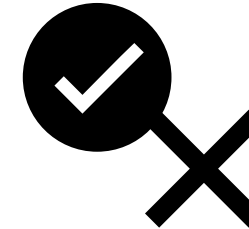


CPMpy

Essence
prime

Constraint Acquisition

Users provide or classify (non-)solutions



From Natural Language

Textual descriptions of combinatorial problems

Making Constraint Solving More Accessible

Modelling Tools

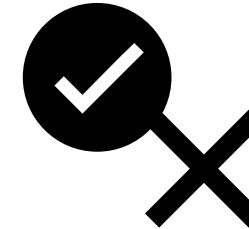


CPMpy

Essence
prime

Constraint Acquisition

Users provide or classify (non-)solutions



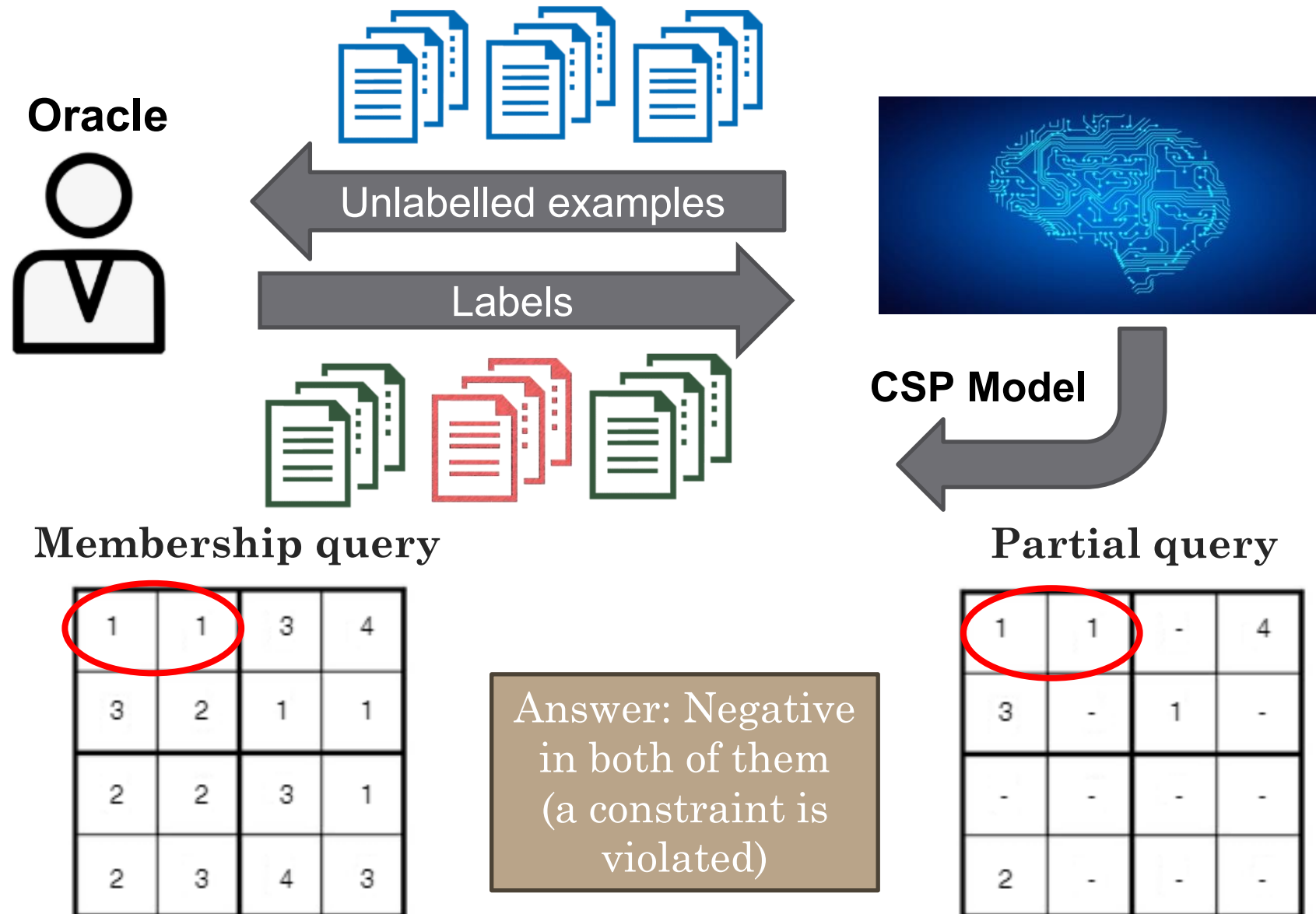
*Modeling is not a
one-shot procedure!*



From Natural Language

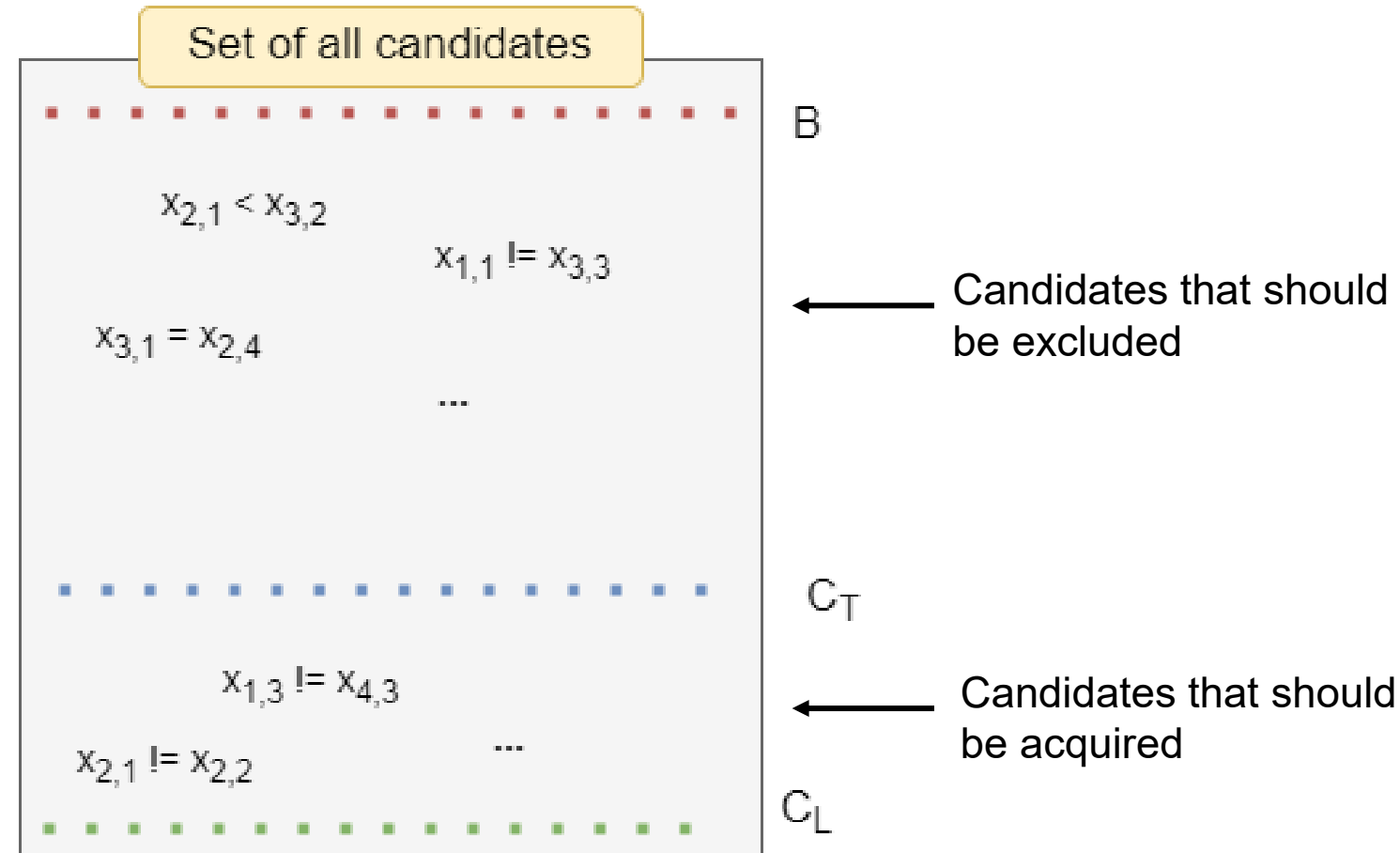
Textual descriptions of combinatorial problems

Interactive Constraint Acquisition (CA)



Interactive CA

- **B**: set of (remaining) candidate constraints
- **C_T**: target set of constraints
- **C_L**: learned set of constraints

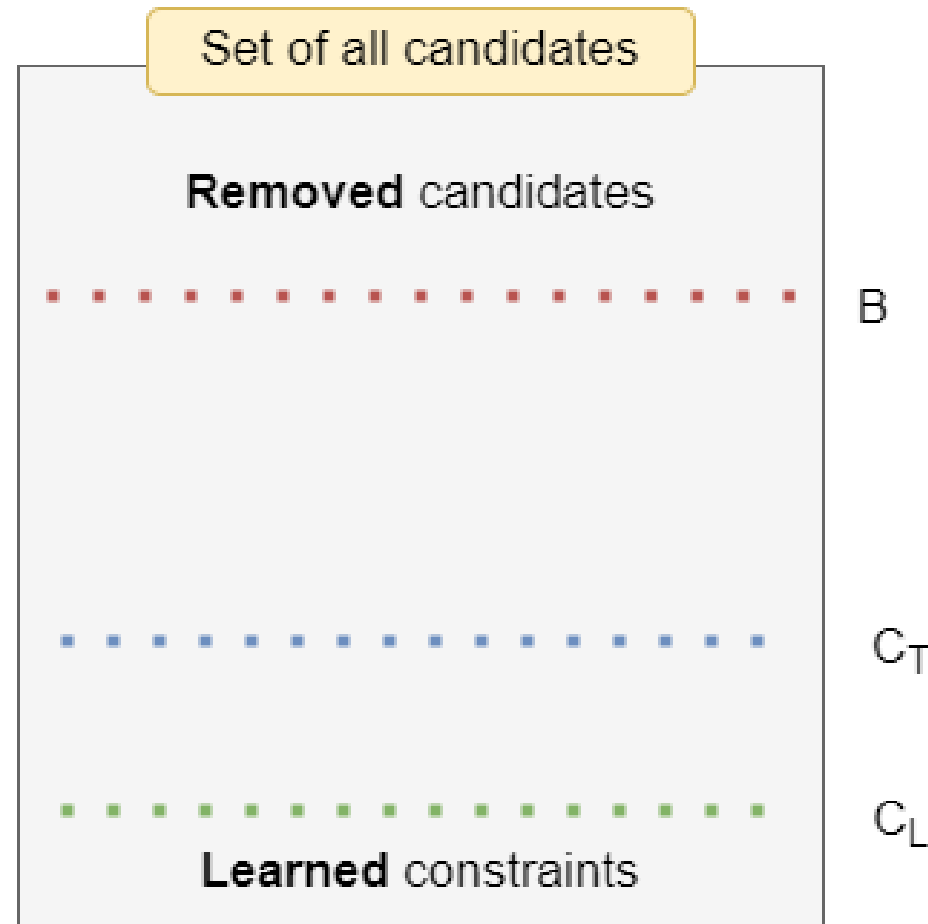


Interactive CA

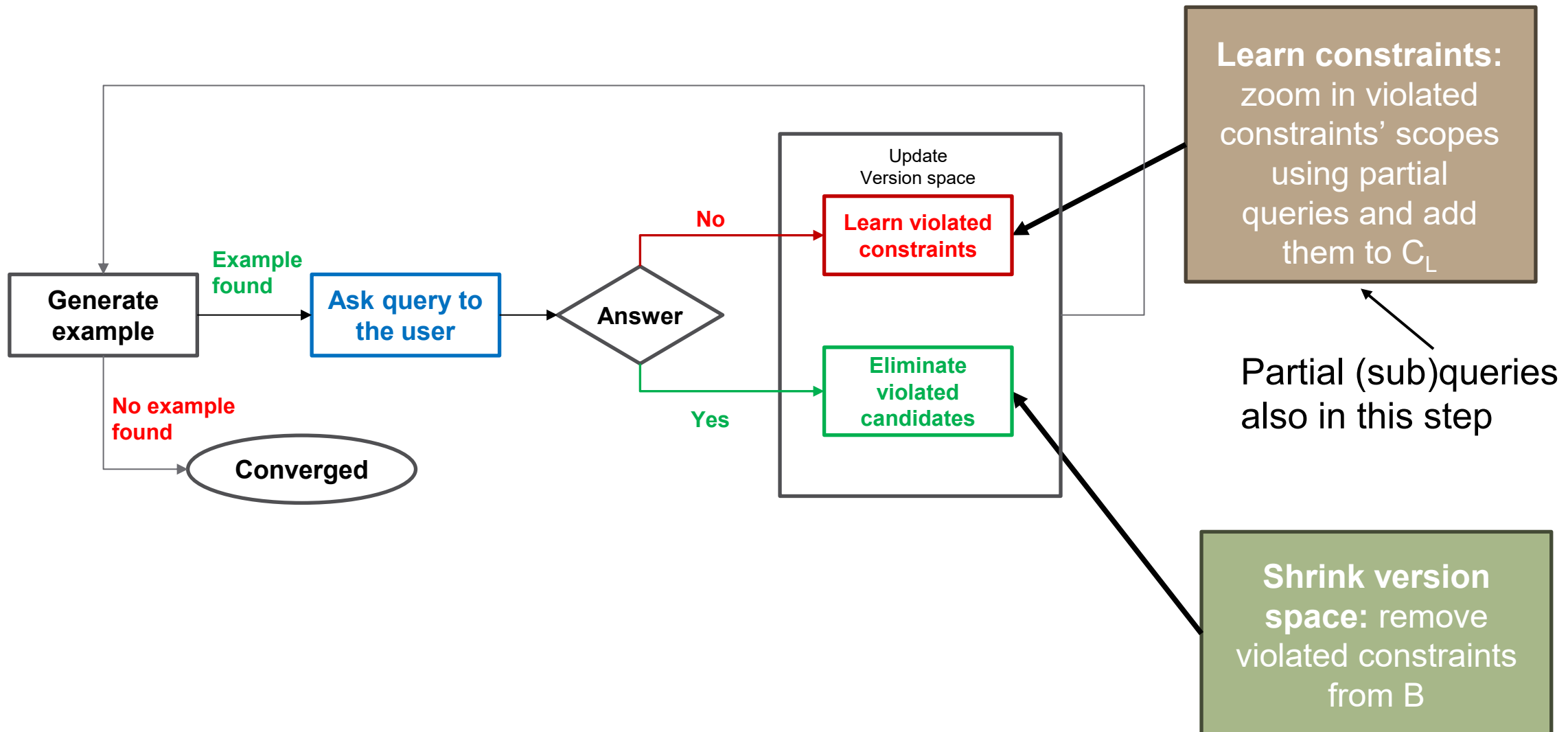
During the learning process:

- Constraints are removed from B
- Constraints are added to C_L

- B : set of (remaining) candidate constraints
- C_T : target set of constraints
- C_L : learned set of constraints



Interactive Constraint Acquisition



Main challenge for human users: Slow learning

Standard interactive CA learns one ground constraint at a time.
Even when many constraints share the same underlying **pattern**.

Example from Exam Timetabling

$course[i,j] \neq course[k,l]$



Broader rule: all courses
should be in different slots

- Learning each constraint separately is slow and repeats work.
- Learning the pattern can **reduce** the number of queries.

Generalization queries in interactive CA

Existing type-based generalization

- Generalization based on user-given variable types (e.g. row, column)
- Ask whether a learned ground constraint should hold for every combination of variables with this sequence of types.

learned ground constraint $\text{cell}(1,1) \neq \text{cell}(1,9)$



Which sequence of types?

- *Not fully expressive*
- *Many failing generalization attempts*

Var1: column1 + Var2: row1 ?

Var1: row1 + Var2: column9 ?

Var1: column1 + Var2: column9

Var1: row1 + Var2: row1 ✓

Constraint Specification (CS)

Capturing **problem-level requirements**

«All variables in the same row should have a different value»

```
Foreach row  $\in$  all_rows:  
  foreach cell1, cell2  $\in$  all_pairs(row):  
    c  $\leftarrow$  cell1  $\neq$  cell2
```

- **What kind of constraint?** $\neq$
- **Which variables?** Same row
- **Which combinations?** All pairs

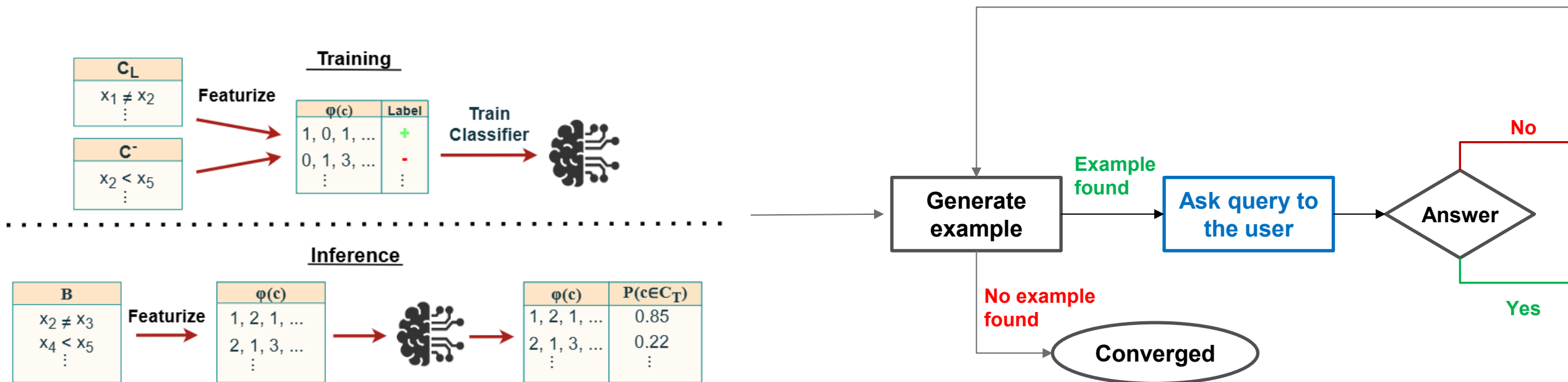
Elements of a CS

```
Foreach Y  $\in$   $\mathcal{P}(V.T)$ :  
  foreach scope  $\in$  combinations(Y, arity(r),  
                                S):  
    c  $\leftarrow$  (rel(c) = r, var(c) = scope)
```

- **Relation** r : What constraint?
- **Partitioning** function G : How to group the variables?
- Sequence **conditions** S : Which combinations in each group?

*Can we **generalize** the learned ground constraints to constraint specifications?*

Existing in CA: Using ML to guide query generation

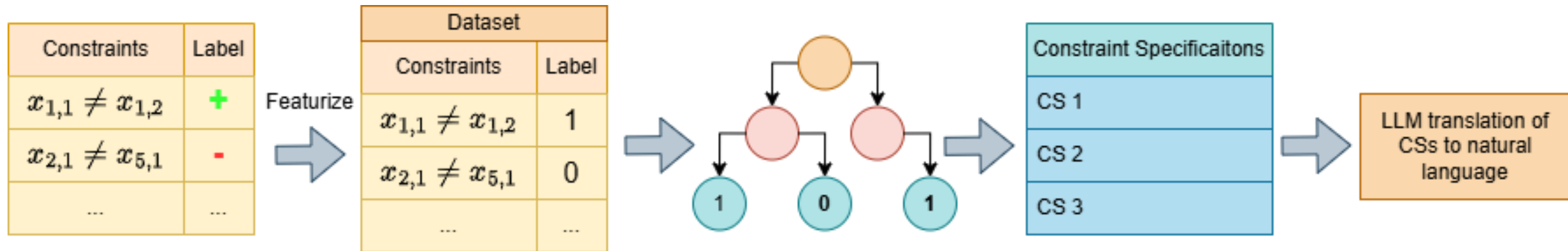


The classifier is learning the patterns to predict constraints

Let's use what the classifier learns to generalize

Framework for extracting constraint specifications using ML

1. Build dataset of constraints during acquisition
2. Train a **rule-based** classifier
3. Extract Constraint Specifications
4. Query the user in natural language



GQ-Gen: Generalization query generation

Extract CSs using ML

Feature Representation of Constraints

Feature group	Example features
Relation	relation name, constants, arity
Partitioning	same row / column / latent dims
Conditioning	index avg, min/max, distances
Label	constraint is in target C_T or not

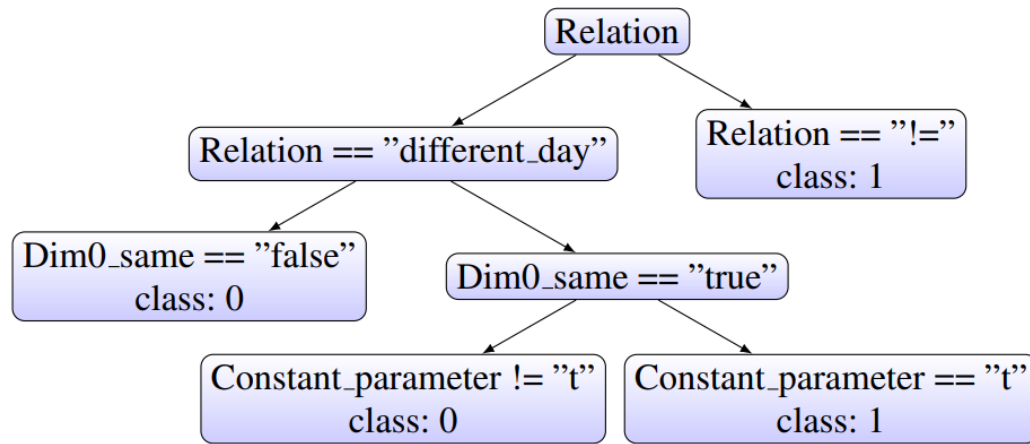
Feature groups: building blocks for extracting CSs from *positive* decision rules.

Extracting CSs

1. *Get the positive decision rule*
2. *Check what features it involves*
3. *Build the CS by putting the features in their respective group*

Example – Exam Timetabling

1. Decision Tree learned



2. Decision Rules

```
r1: Relation == "different_day"
& Dim0_same == "false"
then 0
r2: Relation == "different_day"
& Dim0_same == "true"
& Constant_parameter != "t"
then 0
r3: Relation == "different_day"
& Dim0_same == "true"
& Constant_parameter == "t"
then 1
r4: Relation == "!=" then 1
```

3. Extracted Constraint Specifications

```
Foreach row  $\in$  all_rows:
  Foreach scope  $\in$  all_pairs(row):
     $c \leftarrow (\text{rel}(c) = \text{"different\_day(t)"},$ 
                $\text{var}(c) = \text{scope})$ 
```

```
Foreach scope  $\in$  all_pairs(V):
   $c \leftarrow (\text{rel}(c) = \text{"!="}, \text{var}(c) = \text{scope})$ 
```

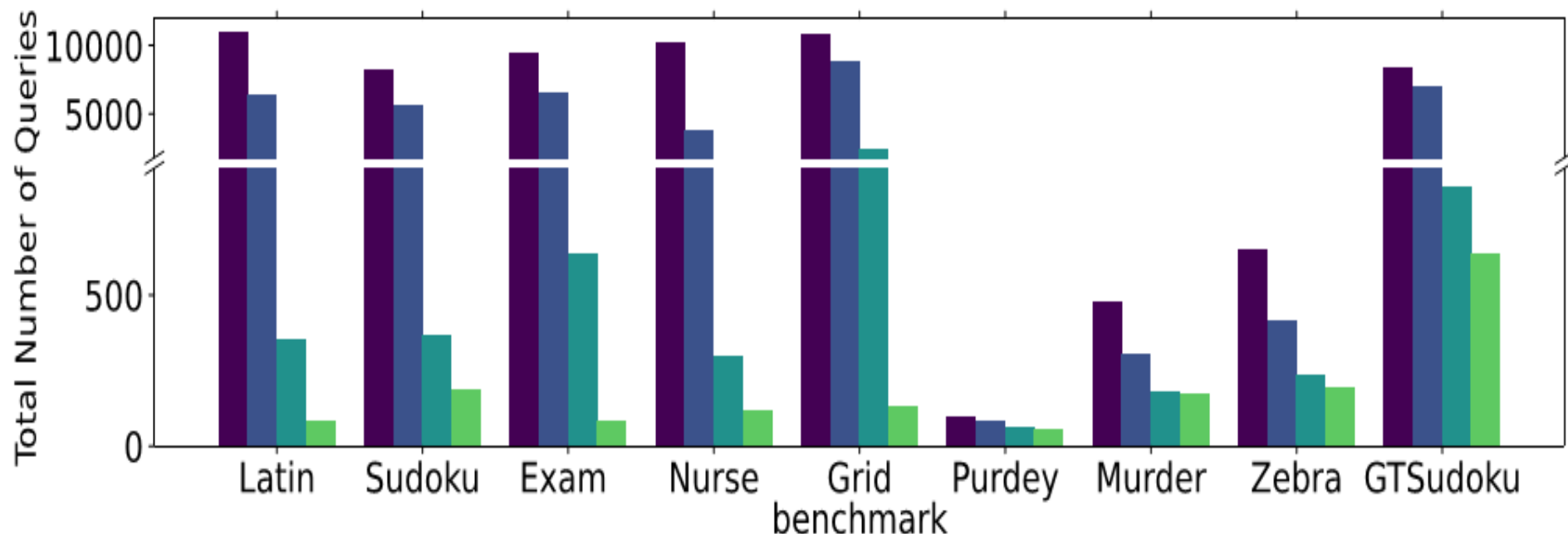
4. Natural language queries

Within each semester, should every pair of exams be scheduled on different days?

In the entire timetable, should every pair of exams be assigned to different slots?

Results - # of queries

■ QuAcq ■ QuAcq + Mine&Ask ■ QuAcq + GenAcq ■ QuAcq + GQ-Gen (ours)



Compared against

1. QuAcq (No generalization)
2. GenAcq (type-based generalization, given types)
3. Mine&Ask (detecting types for generalization)

9 benchmarks:

Fully structured

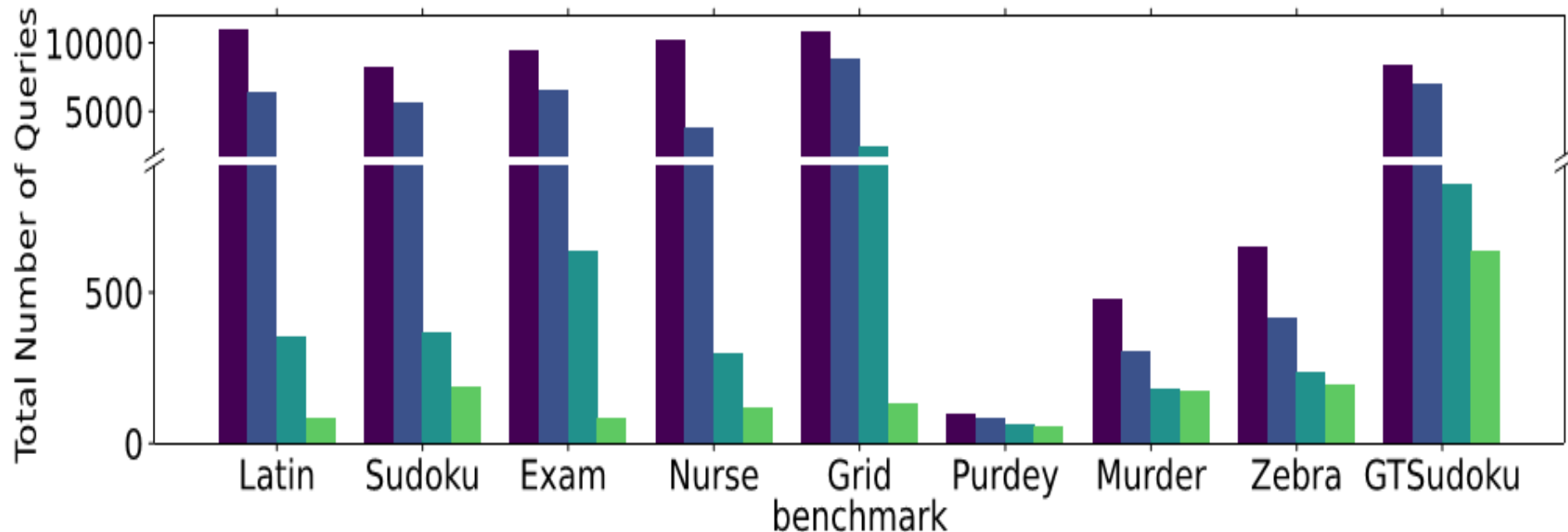
1. **Sudoku**
2. **Latin** squares
3. **Exam** timetabling
4. **Nurse** rostering
5. Distance-based **Grid** colouring

Partially structured

1. Purdey
2. Murder
3. Zebra
4. GTSudoku

Results - # of queries

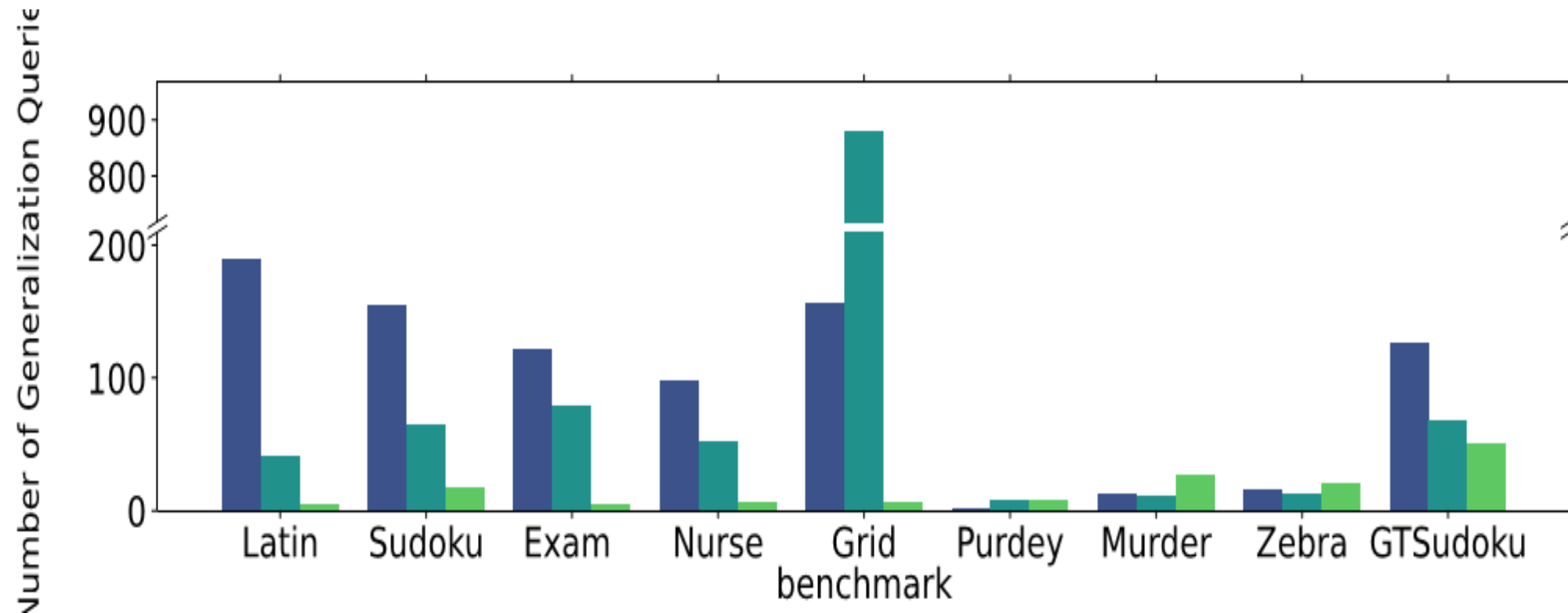
■ QuAcq ■ QuAcq + Mine&Ask ■ QuAcq + GenAcq ■ QuAcq + GQ-Gen (ours)



Up to 99% fewer
queries vs QuAcq

Up to 95% fewer
queries vs GenAcq

Results - # of Generalization Queries



of generalization
queries stays low!

Scalability

Scales much better
than existing
interactive CA
methods

Additional queries in
larger problem sizes
are (mostly) queries
to learn initial
constraints before
generalization

All Queries, Exam Timetabling

<i>Nr. Constraints:</i>	77	143	218	348	488	716	959	1317	1699	2231
QUACQ	377	734	1248	2046	2916	4338	5935	7994	10289	13108
QUACQ + GENACQ	101	149	191	233	385	374	542	557	818	826
QUACQ + GQ-GEN	32	35	37	37	46	60	52	79	66	73

All Queries, Nurse Rostering

<i>Nr. Constraints:</i>	53	171	216	394	476	640	722	804	1275	1854
QUACQ	403	1671	2338	4210	5368	8022	9273	10546	>15K	>15K
QUACQ + GENACQ	75	125	156	182	222	308	356	394	423	442
QUACQ + GQ-GEN	45	115	100	122	232	93	91	202	286	228

All Queries, Grid Coloring

<i>Nr. Constraints:</i>	168	270	396	465	546	627	708	813	918	1140
QUACQ	1229	2359	3831	4621	5628	6702	7868	9325	10852	13828
QUACQ + GENACQ	343	600	930	1138	1349	1494	1751	2017	2305	5016
QUACQ + GQ-GEN	92	162	114	138	215	136	150	119	189	148

User study

- Natural-language formulations of CS-based queries were evaluated by 19 participants.
- Results indicate that users can understand and answer the generated queries quickly and accurately.

79%

High clarity
score

89%

correct
answers

69%

highest
confidence

70%

<30s response

Conclusions

- ❑ Pattern recognition with ML moves from guiding learning to being the **main learning component**.
- ❑ Generalization can make interactive CA substantially more query-efficient and more human-friendly.
- ❑ Next steps:
 - ❑ LLM-based interaction – extending LLMACQ
 - ❑ Generalization without explicit candidate constraints
 - ❑ Data-based generalization

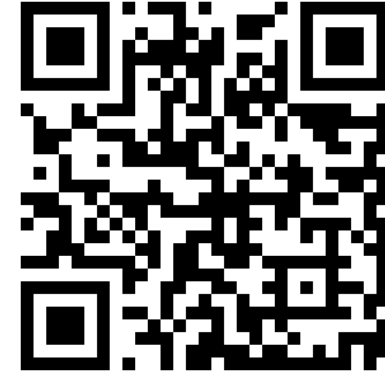
Related material

PyConA



<https://github.com/CPMpy/PyConA>

ML-guided Interactive Constraint Acquisition



<https://doi.org/10.1613/jair.1.19524>

CP + ML Survey - JAIR



<https://jair.org/index.php/jair/article/view/19533>